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Semiconductor substrate processing apparatus

Abstract

A semiconductor substrate processing apparatus includes a substrate transfer module including a chamber having an internal space extending in a first direction within the chamber, at least one pair of first load ports at opposite sides of the chamber, to face in a second direction intersecting the first direction, and configured to rotate and move a substrate carrier, a load lock at a rear surface of the chamber, and a robot arm configured to move in the first direction in the internal space of the chamber, a transfer chamber connected to the load lock of the substrate transfer module, a plurality of processing chambers connected to the transfer chamber, and a transfer arm inside the transfer chamber, and configured to unload the semiconductor substrate from the load lock and to load the semiconductor substrate into at least one of the plurality of processing chambers.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5474410	12/1994	Ozawa	414/217	H01L 21/67766
8985937	12/2014	Fukutomi	414/940	H01L 21/67766
9818633	12/2016	Lill et al.	N/A	N/A
10559483	12/2019	Gould et al.	N/A	N/A
11121015	12/2020	Babbs et al.	N/A	N/A
11121017	12/2020	Hall et al.	N/A	N/A
2002/0192057	12/2001	Meulen	414/217.1	H01L 21/67778
2003/0091410	12/2002	Larson	414/217	H01L 21/67161
2003/0194297	12/2002	Sackett et al.	N/A	N/A
2015/0013910	12/2014	Krupyshev et al.	N/A	N/A
2015/0044001	12/2014	Hofmeister	414/217	H01L 21/67201
2019/0218041	12/2018	Caveney	N/A	N/A
2020/0083071	12/2019	Gould et al.	N/A	N/A
2021/0028044	12/2020	Vellore	N/A	H01L 21/67383
2021/0035830	12/2020	Kim	N/A	H01L 21/67161
2021/0272832	12/2020	Burkhart et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2020012669	12/2019	WO	B25J 11/0095

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims the benefit under 35 U.S.C. 119(a) of Korean Patent Application No. 10-2022-0007281 filed on Jan. 18, 2022 in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference for all purposes.

BACKGROUND

(2) The present inventive concepts relate to semiconductor substrate processing apparatuses.

(3) A semiconductor substrate processing apparatus may include components for transferring a semiconductor substrate in addition to a plurality of processing chambers in which thin films are stacked on a semiconductor substrate, for example, include a load port on which a substrate carrier in which the semiconductor substrate is accommodated is seated, a robot arm unloading the semiconductor substrate from the substrate carrier, a load lock temporarily accommodating the semiconductor substrate input to the plurality of processing chambers, and the like.

SUMMARY

(4) Example embodiments provide a semiconductor substrate processing apparatus having improved processing efficiency per unit area. By reducing the area occupied by these components, processing efficiency of the semiconductor substrate processing apparatus per unit area may be improved, thereby improving the compactness, efficiency, and performance provided by the semiconductor substrate processing apparatus.

(5) According to some example embodiments, a semiconductor substrate processing apparatus includes a substrate transfer module including a chamber having one or more inner surfaces at least partially defining an internal space extending in a first direction within the chamber, at least one pair of first load ports at opposite sides of the chamber, the at least one pair of first load ports facing in a second direction and configured to rotate and move a substrate carrier to couple the substrate carrier accommodating a semiconductor substrate to the chamber, the second direction intersecting the first direction, a load lock at a rear surface of the chamber to be connected to the internal space of the chamber, and a robot arm configured to move in the first direction in the internal space of the chamber to unload the semiconductor substrate from the substrate carrier and load the semiconductor substrate into the load lock or to unload the semiconductor substrate from the load lock and load the semiconductor substrate into the substrate carrier; a transfer chamber connected to the load lock of the substrate transfer module; a plurality of processing chambers connected to the transfer chamber; and a transfer arm inside the transfer chamber, and configured to unload the semiconductor substrate from the load lock and to load the semiconductor substrate into at least one processing chamber of the plurality of processing chambers.

(6) According to some example embodiments, a semiconductor substrate processing apparatus includes a substrate transfer module including a chamber having one or more inner surfaces at least partially defining an internal space extending in a first direction within the chamber, at least one pair of load ports at opposite sides of the chamber, the at least one pair of first load ports facing in a second direction, intersecting the first direction, at least one pair of substrate aligners and at least one pair of buffer devices on opposite side portions of the internal space to overlap the at least one pair of load ports in the first direction, and a robot arm configured to move in the first direction in the internal space of the chamber and configured to transfer a semiconductor substrate to the at least one pair of load ports, the at least one pair of substrate aligners and the at least one pair of buffer devices; a transfer chamber connected to the substrate transfer module; a plurality of processing chambers connected to the transfer chamber; and a transfer arm inside the transfer

chamber and configured to load the semiconductor substrate into at least one of the plurality of processing chambers.

(7) According to some example embodiments, a semiconductor substrate processing apparatus includes a substrate transfer module including a chamber having a front surface and a rear surface opposing each other, at least one pair of load ports at opposite sides of the chamber, at least one pair of buffer devices in the chamber to overlap the at least one pair of load ports in a vertical direction, a load lock on the rear surface of the chamber, and a robot arm configured to move in the vertical direction in the chamber and configured to transfer a semiconductor substrate to the at least one pair of load ports, the at least one pair of buffer devices, and the load lock. The front surface of the chamber includes an entrance configured to enable the robot arm and the at least one pair of buffer devices to be loaded and unloaded through the entrance.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The above and other aspects, features, and advantages of the present inventive concepts will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1A is a perspective view illustrating a semiconductor substrate processing apparatus according to some example embodiments;
- (3) FIG. 1B is a plan view of the semiconductor substrate processing apparatus of FIG. 1A according to some example embodiments;
- (4) FIG. 1C is a front view of the semiconductor substrate processing apparatus of FIG. 1A according to some example embodiments;
- (5) FIG. 2 is a plan view illustrating a semiconductor substrate processing apparatus according to some example embodiments;
- (6) FIG. 3A is a plan view illustrating a processing chamber in which semiconductor substrate processing apparatuses of some example embodiments are disposed;
- (7) FIGS. 3B and 3C are plan views each illustrating a processing chamber in which semiconductor substrate processing apparatuses according to some example embodiments are disposed;
- (8) FIG. 4 is a perspective view illustrating a semiconductor substrate processing apparatus according to some example embodiments;
- (9) FIGS. 5A, 5B, 5C, 5D, and 5E are front views illustrating a process of processing a semiconductor substrate using an apparatus for processing a semiconductor substrate according to some example embodiments;
- (10) FIGS. 6A, 6B, and 6C are front views illustrating a process of processing a semiconductor substrate using an apparatus for processing a semiconductor substrate according to some example embodiments;
- (11) FIG. 7 is a front view illustrating a semiconductor substrate processing apparatus according to some example embodiments;
- (12) FIGS. 8A, 8B, and 8C are front views illustrating a process of processing a semiconductor substrate using an apparatus for processing a semiconductor substrate according to some example embodiments; and
- (13) FIGS. 9A and 9B are perspective views illustrating a semiconductor substrate processing apparatus according to some example embodiments.

DETAILED DESCRIPTION

- (14) Hereinafter, some example embodiments will be described in detail with reference to the accompanying drawings.
- (15) It will be understood that when an element such as a layer, film, region, or substrate is referred

to as being “on” another element, it may be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present. It will further be understood that when an element is referred to as being “on” another element, it may be above or beneath or adjacent (e.g., horizontally adjacent) to the other element.

(16) It will be understood that elements and/or properties thereof (e.g., structures, surfaces, directions, or the like), which may be referred to as being “perpendicular,” “parallel,” “coplanar,” or the like with regard to other elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) may be “perpendicular,” “parallel,” “coplanar,” or the like or may be “substantially perpendicular,” “substantially parallel,” “substantially coplanar,” respectively, with regard to the other elements and/or properties thereof.

(17) Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially perpendicular” with regard to other elements and/or properties thereof will be understood to be “perpendicular” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or angle from “perpendicular,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a tolerance of $\pm 10\%$).

(18) Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially parallel” with regard to other elements and/or properties thereof will be understood to be “parallel” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or angle from “parallel,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a tolerance of $\pm 10\%$).

(19) Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially coplanar” with regard to other elements and/or properties thereof will be understood to be “coplanar” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or angle from “coplanar,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a tolerance of $\pm 10\%$)).

(20) It will be understood that elements and/or properties thereof may be recited herein as being “the same” or “equal” as other elements, and it will be further understood that elements and/or properties thereof recited herein as being “identical” to, “the same” as, or “equal” to other elements may be “identical” to, “the same” as, or “equal” to or “substantially identical” to, “substantially the same” as or “substantially equal” to the other elements and/or properties thereof. Elements and/or properties thereof that are “substantially identical” to, “substantially the same” as or “substantially equal” to other elements and/or properties thereof will be understood to include elements and/or properties thereof that are identical to, the same as, or equal to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances. Elements and/or properties thereof that are identical or substantially identical to and/or the same or substantially the same as other elements and/or properties thereof may be structurally the same or substantially the same, functionally the same or substantially the same, and/or compositionally the same or substantially the same.

(21) It will be understood that elements and/or properties thereof described herein as being “substantially” the same and/or identical encompasses elements and/or properties thereof that have a relative difference in magnitude that is equal to or less than 10%. Further, regardless of whether elements and/or properties thereof are modified as “substantially,” it will be understood that these elements and/or properties thereof should be construed as including a manufacturing or operational tolerance (e.g., $+10\%$) around the stated elements and/or properties thereof.

(22) When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value include a tolerance of $\pm 10\%$

around the stated numerical value. When ranges are specified, the range includes all values therebetween such as increments of 0.1%.

(23) While the term “same,” “equal” or “identical” may be used in description of some example embodiments, it should be understood that some imprecisions may exist. Thus, when one element is referred to as being the same as another element, it should be understood that an element or a value is the same as another element within a desired manufacturing or operational tolerance range (e.g., $\pm 10\%$).

(24) When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value includes a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated numerical value. Moreover, when the words “about” and “substantially” are used in connection with geometric shapes, it is intended that precision of the geometric shape is not required but that latitude for the shape is within the scope of the disclosure. Further, regardless of whether numerical values or shapes are modified as “about” or “substantially,” it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated numerical values or shapes. When ranges are specified, the range includes all values therebetween such as increments of 0.1%.

(25) As described herein, when an operation is described to be performed “by” performing additional operations, it will be understood that the operation may be performed “based on” the additional operations, which may include performing said additional operations alone or in combination with other further additional operations.

(26) As described herein, an element that is described to be “spaced apart” from another element, in general and/or in a particular direction (e.g., vertically spaced apart, laterally spaced apart, etc.) and/or described to be “separated from” the other element, may be understood to be isolated from direct contact with the other element, in general and/or in the particular direction (e.g., isolated from direct contact with the other element in a vertical direction, isolated from direct contact with the other element in a lateral or horizontal direction, etc.). Similarly, elements that are described to be “spaced apart” from each other, in general and/or in a particular direction (e.g., vertically spaced apart, laterally spaced apart, etc.) and/or are described to be “separated” from each other, may be understood to be isolated from direct contact with each other, in general and/or in the particular direction (e.g., isolated from direct contact with each other in a vertical direction, isolated from direct contact with each other in a lateral or horizontal direction, etc.).

(27) FIG. 1A is a perspective view illustrating a semiconductor substrate processing apparatus **1000** according to some example embodiments, FIG. 1B is a plan view of the semiconductor substrate processing apparatus **1000** of FIG. 1A according to some example embodiments, and FIG. 1C is a front view of the semiconductor substrate processing apparatus **1000** of FIG. 1A according to some example embodiments.

(28) Referring to FIGS. 1A to 1C, the semiconductor substrate processing apparatus **1000** according to some example embodiments may include a substrate transfer module **100**, a transfer chamber **200**, a transfer arm **250**, and a processing chamber **300**.

(29) The substrate transfer module **100** may include a chamber **110**, a load port **120**, a load lock **130**, a substrate aligner **140**, a buffer device **150**, a robot arm **160**, and an air supply device **170**. The substrate transfer module **100** may be configured to unload a semiconductor substrate S to be provided in the process from a substrate carrier C, to align the semiconductor substrate S and transfer the semiconductor substrate to the transfer chamber **200**. In addition, the substrate transfer module **100** may be configured to receive the semiconductor substrate S having been completed in the process from the transfer chamber **200**, temporarily store the semiconductor substrate S, and load the semiconductor substrate S into the substrate carrier C. According to some example embodiments, the load port **120** is stacked on the chamber **110**, and the robot arm **160** for transferring the semiconductor substrate S is configured to move in the vertical direction (Z-

direction) within the chamber **110**, thereby reducing an installation area (e.g., an area occupied in the X-Y plane) of the substrate transfer module **100** and improving processing efficiency of the semiconductor substrate processing apparatus **1000** per unit area, thereby improving the compactness, efficiency, and performance provided by the semiconductor substrate processing apparatus **1000**.

(30) The chamber **110** may have a front surface FS and a rear surface BS opposing each other and may include side surfaces SS opposing each other and may include a top surface TS, and an internal space (also described herein as internal space **110i** as shown in at least FIGS. **1B** and **1C**) extending in the first direction (Z-direction). The chamber **110** may be understood to be a structure having one or more inner surfaces that at least partially define the internal space **110i** extending in the first direction (Z-direction), where the one or more inner surfaces may include one or more of the front surface FS, the rear surface BS, one or more of the side surfaces SS, the lower surface LS, any combination thereof, or the like. The internal space of the chamber **110** may provide a movement path of the semiconductor substrate S which is unloaded from the substrate carrier C in a vertical direction (e.g., Z-direction) and a horizontal direction (e.g., a direction in which the X-Y plane extends). In addition, the internal space of the chamber **110** may provide a space in which the substrate aligner **140**, the buffer device **150**, the robot arm **160**, the air supply device **170**, and the like are disposed. In addition, a control module **190** for controlling driving of the load port **120**, the substrate aligner **140**, and the robot arm **160**, controlling any part of the semiconductor substrate processing apparatus **1000**, any combination thereof, or the like, may be disposed inside the chamber **110**. In the drawing, the control module **190** is disposed below the buffer device **150**, but the configuration is not limited thereto.

(31) As described herein, any apparatuses, devices, systems, blocks, modules, units, controllers, circuits, apparatus, and/or portions thereof according to any of some example embodiments (including, without limitation, any of the example embodiments of the semiconductor substrate processing apparatus **1000**, control module **190**, semiconductor substrate processing apparatus **1000A**, semiconductor substrate processing apparatus **1000B**, semiconductor substrate processing apparatus **1000C**, semiconductor substrate processing apparatus **1**, semiconductor substrate processing apparatus **1'**, semiconductor substrate processing apparatus **1000a**, semiconductor substrate processing apparatus **1000b**, any portion thereof, or the like) may include, may be included in, and/or may be implemented by one or more instances of processing circuitry such as hardware including logic circuits; a hardware/software combination such as a processor executing software; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a graphics processing unit (GPU), an application processor (AP), a digital signal processor (DSP), a microcomputer, a field programmable gate array (FPGA), and programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), a neural network processing unit (NPU), an Electronic Control Unit (ECU), an Image Signal Processor (ISP), and the like. In some example embodiments, the processing circuitry may include a non-transitory computer readable storage device (e.g., a memory), for example a solid state drive (SSD), storing a program of instructions, and a processor (e.g., CPU) configured to execute the program of instructions to implement the functionality and/or methods performed by some or all of any apparatuses, devices, systems, blocks, modules, units, controllers, circuits, apparatuses, and/or portions thereof according to any of some example embodiments, and/or any portions thereof, including for example some or all operations of any of the methods and/or processes shown in any of the drawings.

(32) The load port **120** may be configured to support and move the substrate carrier C accommodating the semiconductor substrate S to be coupled to the chamber **110**. To couple the substrate carrier C to the chamber **110**, the load port **120** may be configured to move the substrate carrier C in the horizontal direction (e.g., X-direction) or rotate the substrate carrier C about an axis in the vertical direction (e.g., Z-direction). As an example, the load port **120** may include a transfer

rail **123** extending towards the chamber **110**, and a transfer tray **125** configured to move along the transfer rail **123** and rotate about an axis extending in the first direction (Z-direction). The load port **120** may rotate the substrate carrier C so that a door of the substrate carrier C mounted on the transfer tray **125** faces the chamber **110**. The door of the substrate carrier C may be opened and closed by being coupled to a gate of one side of the chamber **110**.

(33) The load ports **120** may be respectively disposed on both sides (e.g., opposite sides) of the chamber **110**. For example, the load port **120** may include at least a pair of load ports **120** respectively disposed on both sides (e.g., opposite sides) of the chamber **110** to face in the second direction (X-direction) intersecting the first direction (Z-direction) (e.g., to face each other in the second direction). In addition, the load port **120** may be disposed to overlap at least a portion of the chamber **110**. For example, the load port **120** may be disposed adjacent to a side surface between the front surface FS and the rear surface BS of the chamber **110**, and may overlap at least a portion of the chamber **110** in the vertical direction (e.g., Z-direction). In this case, the substrate aligner **140** and/or the buffer device **150** may be disposed in the internal space of the chamber **110** overlapping the load port **120**, and the robot arm **160** may be configured to move in a vertical direction (e.g., Z-direction) in the internal space of the chamber **110** to load or unload the semiconductor substrate S into the load port **120**, the substrate aligner **140** and the buffer device **150**. According to some example embodiments, since the load port **120** is not disposed on the front surface FS of the chamber **110**, the area of the substrate transfer module **100** (e.g., area in the X-Y plane) may be reduced, and processing efficiency per unit area of the semiconductor substrate processing apparatus **1000** may be improved, thereby improving the compactness, efficiency, and performance provided by the semiconductor substrate processing apparatus **1000**.

(34) The load lock **130** may be disposed on the rear surface BS of the chamber **110** to be connected to (e.g., exposed to, open to, etc.) the internal space of the chamber **110**. The load lock **130** may spatially connect the chamber **110** and the transfer chamber **200**. For example, the load lock **130** may be configured to receive the semiconductor substrate S transferred from the substrate transfer module **100** to the transfer chamber **200** or transferred from the transfer chamber **200** to the substrate transfer module **100**, and to temporarily store the semiconductor substrate S. In addition, a gate (not illustrated) that allows or blocks the entry and exit of the semiconductor substrate S may be disposed between the load lock **130** and the chamber **110**, and a gate (not illustrated) for allowing or blocking the entry and exit of the semiconductor substrate S may be disposed between the load lock **130** and the transfer chamber **200**. The load lock **130** may include a loading load lock and an unloading load lock. However, according to some example embodiments, the number of load locks **130** may be variously changed, for example, the load lock **130** may be provided as only one load lock.

(35) The load lock **130** may be configured to control a pressure of an internal region inside the load lock **130** and in which the semiconductor substrate S is accommodated. For example, the load lock **130** may include one or more inner surfaces at least partially defining an internal region inside the load lock **130**, the load lock **130** may be configured to accommodate the semiconductor substrate S within the internal region, and the load lock **130** may be configured to control a pressure of the internal region. For example, when the semiconductor substrate S is transferred to and from the transfer chamber **200**, the pressure inside the load lock **130** may be controlled to have substantially the same vacuum pressure as that of the transfer chamber **200**. Also, when the semiconductor substrate S is transferred to and from the substrate transfer module **100**, the pressure inside the load lock **130** may be controlled to have substantially the same atmospheric pressure as that of the transfer chamber **200**. The load lock **130** may function as a cooling chamber for cooling the temporarily stored semiconductor substrate S.

(36) The substrate aligner **140** may be disposed in the internal space of the chamber **110** to align the semiconductor substrate S. For example, the substrate aligner **140** may include at least one pair of substrate aligners **140** respectively disposed on both sides of the internal space of the chamber **110**

to overlap with at least one of the at least one pair of load ports **120** in the first direction (Z-direction). The substrate aligner **140** may include a spinner that rotates the semiconductor substrate S so that the semiconductor substrate S faces a preset direction. For example, the substrate aligner **140** may be configured to sense a notch formed in the semiconductor substrate S and rotate the semiconductor substrate S based on the sensed position of the notch. The position of the substrate aligner **140** is not particularly limited, but may be disposed adjacent to the buffer device **150** in consideration of the movement path of the robot arm **160**.

(37) The buffer device **150** may be disposed in the internal space of the chamber **110** and configured to temporarily store the semiconductor substrate S. For example, the buffer device **150** may include at least one pair of buffer devices **150** respectively disposed on both sides of the internal space of the chamber **110** so as to overlap with at least one of the at least one pair of load ports **120** in the first direction (Z-direction). For example, the robot arm **160** may be configured to unload the semiconductor substrate S from the at least one pair of buffer devices **150** and load the semiconductor substrate S into the load lock **130**, or to unload the semiconductor substrate S from the load lock **130** and load the semiconductor substrate S into at least one pair of buffer devices **150**.

(38) In the internal space of the chamber **110**, the robot arm **160** may be configured to transfer the semiconductor substrate S to at least one pair of load ports **120**, at least one pair of substrate aligners **140**, and at least one pair of buffer devices **150**. For example, the robot arm **160** may be configured to move in the first direction (Z-direction), and to unload the semiconductor substrate S from the substrate carrier C mounted on the load port **120** and load the semiconductor substrate S into the substrate aligner **140**, the buffer device **150**, or the load lock **130**, or to unload the semiconductor substrate S from the load lock **130** and load the semiconductor substrate S into the buffer device **150** or the substrate carrier C. The robot arm **160** may be configured to move inside the chamber **110** along a guide rail extending in the first direction (Z-direction).

(39) The air supply device **170** may be disposed in the internal space of the chamber **110** and may be configured to purify the internal space of the chamber **110**. The air supply device **170** may include a fan-filter unit (FFU) including a fan, an intake duct, an exhaust duct, a filter, and the like. In addition, the air supply device **170** may be configured to control the humidity inside the chamber **110**. As an example, the air supply device **170** may include a first air supply device **170a** disposed in the upper portion of the internal space of the chamber **110** and configured to supply air in the vertical direction (Z-direction) to the internal space, and a second air supply device **170b** disposed in the side of the internal space of the chamber **110** (e.g., one side of the substrate aligner **140** and/or one side of the buffer device **150**), and supplying air to the internal space in a horizontal direction (X-direction). The air supply device **170** may form an airflow in a vertical direction (Z-direction) or/and a horizontal direction (X-direction) to reduce particle adhesion on the semiconductor substrate S.

(40) The transfer chamber **200** may be configured to transfer the semiconductor substrate S, between the substrate transfer module **100** and the processing chamber **300**. The transfer chamber **200** may be connected to the load lock **130** of the substrate transfer module **100**. The transfer chamber **200** may be spatially connected to the substrate transfer module **100** (or the load lock **130**) and the processing chamber **300**. The transfer chamber **200** may thus be configured to establish spatial communication between the substrate transfer module **100** and the processing chamber **300** and enable transfer of the semiconductor substrate S between the substrate transfer module **100** and the processing chamber **300**. For example, the transfer chamber **200** may be configured to transfer the semiconductor substrate S received from the substrate transfer module **100** to the processing chamber **300**, or to transfer the semiconductor substrate S received from the processing chamber **300** to the substrate transfer module **100**. Between the transfer chamber **200** and the substrate transfer module **100** (or the load lock **130**), a gate (not illustrated) that allows or blocks the entry and exit of the semiconductor substrate S may be disposed, and a gate (not illustrated) for allowing

or blocking the entry and exit of the semiconductor substrate S may be disposed between the transfer chamber **200** and the processing chamber **300**. The transfer arm **250** configured to load or unload the semiconductor substrate S when the gate (not illustrated) is opened may be disposed inside the transfer chamber **200**. The pressure inside the transfer chamber **200** may be controlled to have substantially the same vacuum pressure as that of the processing chamber **300**.

(41) The transfer arm **250** may be disposed inside the transfer chamber **200** and configured to unload the semiconductor substrate S from the substrate transfer module **100** (or the load lock **130**) and load the semiconductor substrate S into at least one of the plurality of processing chambers **300**. For example, the transfer arm **250** may be configured to move in the horizontal direction (the direction in which the X-Y plane extends) inside the transfer chamber **200**, to unload the semiconductor substrate S from the substrate transfer module **100** (or the load lock **130**) and load the semiconductor substrates S into the processing chamber **300**, or to unload the semiconductor substrate S having been completed in the process from the processing chamber **300** and load the semiconductor substrates S into the substrate transfer module **100** (or the load lock **130**). The transfer arm **250** may be a robot arm configured to handle the semiconductor substrate S in a vacuum state.

(42) The processing chamber **300** may be an apparatus for processing the semiconductor substrate S received from the transfer chamber **200**. The processing chamber **300** may be provided as a plurality of processing chambers **300** connected to the transfer chamber **200**. The processing chamber **300** may include a deposition chamber, such as, for example, a Chemical Vapor Deposition (CVD) chamber, a Physical Vapor Deposition (PVD) chamber and an Atomic Layer Deposition (ALD) chamber, an oxidation chamber, a thermal treatment chamber, a cleaning chamber, an etch chamber, and the like. For example, a first processing box **310** may be disposed in an upper portion of the processing chamber **300**. The first processing box **310** may store components or gases for plasma processing. In addition, second processing boxes **320** may be disposed in a lower portion of the processing chamber **300**. The second processing boxes **320** may accommodate components for processing the semiconductor substrate S in the processing chamber **300**, for example, plasma equipment.

(43) On the other hand, a carrier transfer apparatus **400** may be disposed on the substrate transfer module **100**. The carrier transfer apparatus **400** may be configured to transport the substrate carrier C. For example, the carrier transfer apparatus **400** may transfer the substrate carrier C in which the semiconductor substrate S to be provided to the process is accommodated to the substrate transfer module **100**, or may receive the substrate carrier C in which the process-completed semiconductor substrate S is accommodated, from the substrate transfer module **100**, and transfer the same to an apparatus for a subsequent process. The substrate carrier C may include a front opening unified pod (FOUP) that accommodates a plurality of semiconductor substrates S (e.g., a semiconductor wafer).

(44) The carrier transfer apparatus **400** may include a carrier transfer line **410** and a carrier transfer robot **420**. The carrier transfer line **410** may be disposed above the substrate transfer module **100** and provide a horizontal movement path of the carrier transfer robot **420**. For example, the carrier transfer line **410** may extend in a horizontal direction (e.g., a direction in which the X-Y plane extends) above the substrate transfer module **100**.

(45) The carrier transfer robot **420** may move the substrate carrier C along a path provided by the carrier transfer line **410**. In addition, the carrier transfer robot **420** may be configured to move in the vertical direction to provide the substrate carrier C to the load port **120** of the substrate transfer module **100** or to pick up the substrate carrier C from the load port **120**. The carrier transfer robot **420** may be a robot arm configured to grip the substrate carrier C.

(46) FIG. 2 is a plan view illustrating a semiconductor substrate processing apparatus **1** according to some example embodiments. FIG. 2 illustrates a planar shape of the semiconductor substrate processing apparatus **1** corresponding to FIG. 1B.

(47) Referring to FIG. 2, the semiconductor substrate processing apparatus **1** of some example

embodiments may include a substrate transfer module **10** including a chamber **11**, a load port **12** and a load lock **13**, a transfer chamber **20**, a transfer arm **25**, and a processing chamber **30**. In this case, the load port **12** may be disposed so as not to overlap the chamber **11** in the vertical direction (Z-direction). For example, the load port **12** may be disposed on the front surface FS of the chamber **11**, and the load lock **13** may be disposed on the rear surface BS of the chamber **11**. Accordingly, the Y-direction length of the semiconductor substrate processing apparatus **1** of some example embodiments may be greater than the Y-direction length of the semiconductor substrate processing apparatus **1000** of some example embodiments, including the example embodiments described with reference to FIGS. **1A** to **1C**. Also, the semiconductor substrate processing apparatus **1** of some example embodiments may have a larger installation area than the semiconductor substrate processing apparatus **1000** of some example embodiments. In this case, the “installation area” may be understood as an area occupied by the semiconductor substrate processing apparatus **1000** in the X-Y plane. As a result, compared with the semiconductor substrate processing apparatus **1** of some example embodiments, the processing efficiency of the semiconductor substrate processing apparatus **1000** per unit area in some example embodiments may be improved, thereby improving the compactness, efficiency, and performance provided by the semiconductor substrate processing apparatus **1000**.

(48) FIG. **3A** is a plan view illustrating a processing room **P'** in which the semiconductor substrate processing apparatuses **1** of some example embodiments are disposed, and FIGS. **3B** and **3C** are plan views illustrating a processing room **P** in which semiconductor substrate processing apparatuses **1000a** and **1000b** according to some example embodiments are disposed.

(49) Referring to FIG. **3A**, the processing room **P'** may accommodate a plurality of semiconductor substrate processing apparatuses **1'** according to some example embodiments. The plurality of semiconductor substrate processing apparatuses **1'** may be disposed to be spaced apart from each other in consideration of accessibility for maintenance or the like. For example, on the X-Y plane, the plurality of semiconductor substrate processing apparatuses **1'** may be disposed in the form of a matrix including a row (e.g., ‘2 rows’) of **M** that is an integer of 1 or more and a column (e.g., ‘8 columns’) of **N** that is an integer of 1 or more. Also, the plurality of semiconductor substrate processing apparatuses **1'** may be disposed such that the load ports **12** are aligned on the carrier transfer line **41**. For example, when the carrier transfer line **41** extends in the X-direction, the plurality of semiconductor substrate processing apparatuses **1'** may be disposed to face each other in the Y-direction. In this case, the processing room **P'** may have a first width **L1** in the Y-direction in consideration of the separation distance between the plurality of semiconductor substrate processing apparatuses **1'**.

(50) FIG. **3B** illustrates an inline type semiconductor substrate processing apparatus **1000a** in which processing chambers **300** are arranged along both sides of the transfer chamber **200**, and FIG. **3C** illustrates a cluster type semiconductor substrate processing apparatus **1000b** in which processing chambers **300** are arranged to surround the transfer chamber **200**.

(51) Referring to FIGS. **3B** and **3C**, the processing room **P** may accommodate a plurality of the semiconductor substrate processing apparatuses **1000a** and **1000b** according to some example embodiments. The plurality of semiconductor substrate processing apparatuses **1000a** and **1000b** may be disposed to be spaced apart from each other in consideration of accessibility for maintenance or the like. For example, on the X-Y plane, the plurality of semiconductor substrate processing apparatuses **1000a** and **1000b** may be disposed in the form of a matrix including a row (e.g., ‘2 rows’) of **M** that is an integer of 1 or more and a column (e.g., ‘8 columns’) of **N** that is an integer of 1 or more. Also, the plurality of semiconductor substrate processing apparatuses **1000a** and **1000b** may be disposed such that the load ports **120** are aligned on the carrier transfer line **410**. For example, when the carrier transfer line **410** extends in the X-direction, the plurality of semiconductor substrate processing apparatuses **1000a** and **1000b** may be disposed to face each other in the Y-direction. In this case, the processing room **P** may have a second width **L2** in the Y-

direction in consideration of the separation distance between the plurality of semiconductor substrate processing apparatuses **1000**.

(52) In example embodiments, since the load ports **120** are disposed on both sides of the chamber **110** and at least some of the load ports **120** are disposed to overlap the chamber **110** in the vertical direction (Z-direction), the length of the plurality of semiconductor substrate processing apparatuses **1000a** and **1000b** may be reduced in the Y-direction than that of the semiconductor substrate processing apparatuses **1'** of FIG. 3A. Accordingly, the installation area of the semiconductor substrate processing apparatuses **1000a** and **1000b** in some example embodiments may be smaller than the installation area of the semiconductor substrate processing apparatuses **1'** of some example embodiments. Also, the second width L2 of the processing room P of some example embodiments may be shorter than the first width L1 of the processing room P' of some example embodiments. On the other hand, unlike illustrated in the drawing, depending on the size of the processing room P, the semiconductor substrate processing apparatuses **1000a** and **1000b** according to some example embodiments may be disposed in greater number than the number of the semiconductor substrate processing apparatuses **1'** of some example embodiments. As such, the semiconductor substrate processing apparatuses **1000a** and **1000b** according to some example embodiments may provide the processing room P having superior processing efficiency per unit area than the semiconductor substrate processing apparatuses **1'** of some example embodiments.

(53) FIG. 4 is a perspective view illustrating a semiconductor substrate processing apparatus **1000A** according to some example embodiments.

(54) Referring to FIG. 4, the semiconductor substrate processing apparatus **1000A** according to some example embodiments may further include a second load port **120b** staggered from the first load port **120a**. The second load port **120b** may be disposed staggered from the first load port **120a** to load and unload the substrate carrier C in the vertical direction (Z-direction). For example, the semiconductor substrate processing apparatus **1000A** of some example embodiments may include at least one pair of first load ports **120a** respectively disposed on both sides of the chamber **110** to face each other in the second direction (X-direction), and at least one pair of second load ports **120b** respectively disposed on both sides of the chamber **110** so as not to overlap the at least one pair of first load ports **120a** in the first direction (Z-direction). According to some example embodiments, without an increase in the Y-direction length of the semiconductor substrate processing apparatus **1000A**, four or more substrate carriers C may be mounted, and the processing efficiency of the semiconductor substrate processing apparatus **1000A** per unit area may be further improved. Since the semiconductor substrate processing apparatus **1000A** of some example embodiments has the same or similar characteristics as those described with reference to FIGS. 1A to 1C, overlapping descriptions will be omitted.

(55) FIGS. 5A, 5B, 5C, 5D, and 5E are front views illustrating a process of processing a semiconductor substrate using a semiconductor substrate processing apparatus according to some example embodiments. FIGS. 5A to 5E illustrate a process of unloading the semiconductor substrate S from the substrate carrier C and loading the semiconductor substrate S into the load lock **130** in sequence.

(56) Referring to FIG. 5A, first, the substrate carrier C may be mounted on at least one of the plurality of load ports **120a** and **120b**. For example, the carrier transfer robot **420** moves in the horizontal direction (the direction in which the X-Y plane extends) and the vertical direction (the Z-direction), and may mount the substrate carrier C on at least one of the first load port **120a** and the second load port **120b**. The remaining part of the first load port **120a** and the second load port **120b** may be in a state in which the substrate carrier C on which the semiconductor substrate is loaded or unloaded is mounted.

(57) Referring to FIG. 5B, by operating the load ports **120a** and **120b** on which the substrate carrier C is mounted, the substrate carrier C may be coupled to the chamber **110** of the substrate transfer module **100**. For example, the transfer tray **125** supporting the substrate carrier C rotates about 90

degrees about an axis extending in the vertical direction (Z-direction), and moves along the transfer rail **123** in a horizontal direction (e.g., X-direction) to couple the door of the substrate carrier **C** to the gate of one side of the chamber **110**. Thereafter, the gate of the chamber **110** and the door of the substrate carrier **C** are opened, and the internal space of the substrate carrier **C** may be connected to the internal space of the chamber **110**.

(58) Referring to FIG. 5C, the semiconductor substrate **S** may be unloaded from the substrate carrier **C**. For example, the robot arm **160** moves in the vertical direction (Z-direction) to be adjacent to the substrate carrier **C** and then expands and contracts in the horizontal direction (e.g., the X-direction) to unload the semiconductor substrate **S** from the substrate carrier **C**.

(59) Referring to FIG. 5D, the semiconductor substrate **S** may be loaded into the substrate aligner **140**. For example, the robot arm **160** moves in a vertical direction (Z-direction) to be adjacent to the substrate aligner **140**, and then, may expand and contract in a horizontal direction (e.g., X-direction) to load the semiconductor substrate **S** into the substrate aligner **140**. The substrate aligner **140** may sense a notch formed in one side of the semiconductor substrate **S**, and may rotate the semiconductor substrate **S** in a preset direction based on the sensed position of the notch.

Thereafter, according to some example embodiments, the semiconductor substrate **S** may be unloaded from the substrate aligner **140** and loaded into the buffer device **150**.

(60) Referring to FIG. 5E, the semiconductor substrate **S** may be unloaded from the substrate aligner **140** or the buffer device **150** and loaded into the load lock **130**. For example, the robot arm **160** holding the semiconductor substrate **S** moves in the vertical direction (Z-direction) adjacently to the load lock **130**, and then expands and contracts in the horizontal direction (e.g., the X-direction) to load the semiconductor substrate **S** into the load lock **130**. To load the semiconductor substrate **S**, a gate disposed between the load lock **130** and the chamber **110** may be opened and closed. Thereafter, the pressure inside the load lock **130** is controlled to a vacuum pressure, and a transfer arm (refer to '250' of FIG. 1B) of the transfer chamber (refer to '200' of FIG. 1B) may unload the semiconductor substrate **S** from the load lock **130**. To take out the semiconductor substrate **S**, the gate between the load lock **130** and the transfer chamber **200** may be opened and closed.

(61) FIGS. 6A to 6C are front views illustrating a process of processing a semiconductor substrate using a semiconductor substrate processing apparatus according to some example embodiments. FIGS. 6A to 6C sequentially illustrate a process of unloading the semiconductor substrate **S** from the load lock **130** and loading the same into the substrate carrier **C**.

(62) Referring to FIG. 6A, the semiconductor substrate **S** on which the process has been completed may be unloaded from the load lock **130**. For example, after the robot arm **160** moves in the vertical direction (Z-direction) adjacently to the load lock **130**, the semiconductor substrate **S** may be unloaded from the load lock **130** as the robot arm **160** expands and contracts in the horizontal direction (e.g., the X-direction). In this case, the pressure inside the load lock **130** may be adjusted to atmospheric pressure, which is substantially the same as the pressure in the internal space of the chamber **110**. To take out the semiconductor substrate **S**, a gate disposed between the load lock **130** and the chamber **110** may be opened and closed.

(63) Referring to FIG. 6B, the semiconductor substrate **S** on which the process has been completed may be loaded into the buffer device **150**. For example, after the robot arm **160** holding the semiconductor substrate **S** having been completed in the process moves in the vertical direction (Z-direction) to be adjacent to the buffer device **150**, the semiconductor substrate **S** may be loaded into the buffer device **150** as the robot arm **160** expands and contracts in the horizontal direction (e.g., the X-direction). The buffer device **150** may temporarily store the semiconductor substrate **S** on which the process has been completed.

(64) Referring to FIG. 6C, the wafer substrate **S** may be unloaded from the buffer device **150** and loaded into the substrate carrier **C** mounted on the load ports **120a** and **120b**. For example, the robot arm **160** in which the process has been completed moves in the vertical direction (Z-

direction) adjacent to the substrate carrier C and then expands and contracts in the horizontal direction (e.g., the X-direction) to load the semiconductor substrate S into the substrate carrier C. When the semiconductor substrate S on which the process has been completed is loaded into the substrate carrier C, the gate of the chamber **110** and the door of the substrate carrier C may be closed. Then, the carrier transfer robot **420** moves in the vertical direction (Z-direction) to pick up the substrate carrier C, and moves in the horizontal direction (the direction in which the X-Y plane extends), to transfer the substrate carrier C to a separate device for subsequent processing.

(65) FIG. 7 is a front view illustrating a semiconductor substrate processing apparatus **1000B** according to some example embodiments.

(66) Referring to FIG. 7, a semiconductor substrate processing apparatus **1000B** according to some example embodiments has the same or similar characteristics as those described with reference to FIGS. 1A to 6C, except for further including an upper buffer device **155** inside the substrate transfer module **100**, and thus, a redundant description will be omitted. The substrate transfer module of some example embodiments may include the upper buffer device **155** disposed in an upper portion of the internal space of the chamber **110** and configured to store the semiconductor substrate S. In addition, a carrier storage device **450** may be disposed on one side of the carrier transfer robot **420**. The upper buffer device **155** may be disposed so as not to overlap the movement path of the robot arm **160**. For example, the first air supply device **170a** and the upper buffer device **155** may be disposed in a recess formed toward the rear side of the chamber **110** (refer to FIG. 7). However, according to some example embodiments, the first air supply device **170a** is disposed just above the robot arm **160** similarly to that illustrated in FIG. 1A, and an additional air supply device (not illustrated) may be further disposed on the side or rear surface of the upper buffer device **155**. Similar to the buffer device **150** disposed below the load ports **120**, the upper buffer device **155** may be configured to store a semiconductor substrate S to be provided for a process or a semiconductor substrate S on which a process has been completed. According to some example embodiments, when the buffer device **150** is saturated with the semiconductor substrate S, an accommodating space of the semiconductor substrate S may be additionally secured. Hereinafter, a process of loading the semiconductor substrate S into the upper buffer device **155** in the semiconductor substrate processing apparatus **1000B** of some example embodiments will be described with reference to FIGS. 8A to 8C.

(67) FIGS. 8A to 8C are front views illustrating a process of processing a semiconductor substrate using the semiconductor substrate processing apparatus **1000B** of FIG. 7. FIGS. 8A to 8C illustrate a process of loading the semiconductor substrate S into the upper buffer device **155** in order.

(68) Referring to FIG. 8A, the carrier transfer robot **420** moves in the horizontal direction (the direction in which the X-Y plane extends) and the vertical direction (the Z-direction), and the substrate carrier C may be mounted on at least one of the first load port **120a** and the second load port **120b**. Then, by operating the load ports **120a** and **120b** on which the substrate carrier C is mounted, the substrate carrier C may be coupled to the chamber **110** of the substrate transfer module **100**, and the gate of the chamber **110** and the door of the substrate carrier C may be opened. In addition, a carrier storage device **450** may be disposed on one side of the carrier transfer robot **420**.

(69) Referring to FIG. 8B, the semiconductor substrate S may be unloaded from the substrate carrier C and loaded into the upper buffer device **155**. For example, after the robot arm **160** moves in the vertical direction (Z-direction) adjacent to the substrate carrier C, the semiconductor substrate S may be unloaded from the substrate carrier C by extending and contracting in the horizontal direction (e.g., the X-direction). Then, after the robot arm **160** moves in the vertical direction (Z-direction) adjacent to the upper buffer device **155**, the semiconductor substrate S may be loaded into the upper buffer device **155** by extending and contracting in the horizontal direction (e.g., the X-direction).

(70) Referring to FIG. 8C, the substrate carrier C from which all the semiconductor substrates S are

unloaded may be moved to the carrier storage device **450**. For example, the carrier transfer robot **420** moves in the vertical direction (Z-direction) to pick up the substrate carrier C, and the substrate carrier C may be moved to the carrier storage device **450** by moving in the horizontal direction (the direction in which the X-Y plane extends). The carrier storage device **450** may temporarily store the substrate carrier C from which all the semiconductor substrates S are unloaded. Thereafter, in order to take out the semiconductor substrate S on which the process is completed, the carrier transfer robot **420** may provide the substrate carrier C stored in the carrier storage device **450** to the substrate transfer module **100**.

(71) FIGS. **9A** and **9B** are perspective views illustrating a semiconductor substrate processing apparatus **1000C** according to some example embodiments.

(72) Referring to FIGS. **9A** and **9B**, in the semiconductor substrate processing apparatus **1000C** according to some example embodiments, an entrance for loading and unloading the buffer device **150** and the robot arm **160** may be formed on the front surface of the chamber **110**. According to some example embodiments, at least a pair of load ports **120** are disposed to face each other in the X-direction on both sides of the chamber **110** rather than the front surface (e.g., Y-direction) of the chamber **110**, and thus, processing efficiency of the semiconductor substrate processing apparatus **1000C** per unit area may be improved. Accordingly, the substrate transfer module **100** may be configured to maintain the elements inside the chamber **110** through the front surface (e.g., Y-direction) of the chamber **110**. For example, the buffer device **150** and the robot arm **160** may be installed to slide through the front surface (e.g., Y-direction) of the chamber **110**. In addition, the substrate transfer module **100** of some example embodiments may further include at least one front door configured to open and close the entrance and exit of the buffer device **150** and the robot arm **160**.

(73) As set forth above, according to some example embodiments, by significantly reducing the installation area of the substrate transfer module, a semiconductor substrate processing apparatus having improved processing efficiency per unit area may be provided. While some example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present inventive concepts as defined by the appended claims.

Claims

1. A semiconductor substrate processing apparatus, comprising: a substrate transfer module, the substrate transfer module including a chamber having one or more inner surfaces at least partially defining an internal space extending in a first direction within the chamber, at least one pair of first load ports at opposite sides of the chamber, the at least one pair of load ports facing in a second direction and configured to rotate and move a substrate carrier to couple the substrate carrier accommodating a semiconductor substrate to the chamber, the second direction intersecting the first direction, a load lock at a rear surface of the chamber to be connected to the internal space of the chamber, and a robot arm configured to move in the first direction in the internal space of the chamber to unload the semiconductor substrate from the substrate carrier and load the semiconductor substrate into the load lock, or unload the semiconductor substrate from the load lock and load the semiconductor substrate into the substrate carrier; a transfer chamber connected to the load lock of the substrate transfer module; a plurality of processing chambers connected to the transfer chamber; and a transfer arm inside the transfer chamber, the transfer arm configured to unload the semiconductor substrate from the load lock, and load the semiconductor substrate into at least one processing chamber of the plurality of processing chambers, wherein the load lock is configured to connect the transfer chamber to the internal space of the chamber in a third direction, and wherein the first direction, the second direction, and the third direction are perpendicular to each other.

2. The semiconductor substrate processing apparatus of claim 1, wherein the at least one pair of first load ports overlap at least a portion of the chamber in the first direction.
3. The semiconductor substrate processing apparatus of claim 1, wherein the at least one pair of first load ports include a transfer rail extending toward the chamber, and a transfer tray configured to move along the transfer rail and configured to rotate about a rotation axis extending in the first direction.
4. The semiconductor substrate processing apparatus of claim 3, wherein the at least one pair of first load ports are configured to couple the substrate carrier mounted on the transfer tray to the chamber.
5. The semiconductor substrate processing apparatus of claim 1, wherein the substrate transfer module further includes a first air supply device in an upper portion of the internal space and configured to supply air to the internal space.
6. The semiconductor substrate processing apparatus of claim 1, wherein the substrate transfer module further includes a substrate aligner on a side of the internal space to overlap with at least one of the at least one pair of first load ports in the first direction, the substrate aligner configured to align the semiconductor substrate.
7. The semiconductor substrate processing apparatus of claim 1, wherein the substrate transfer module further includes a buffer device on a side of the internal space to overlap with at least one of the at least one pair of first load ports in the first direction, the buffer device configured to store the semiconductor substrate.
8. The semiconductor substrate processing apparatus of claim 7, wherein the substrate transfer module further includes a second air supply device on one side of the buffer device and configured to supply air to the internal space.
9. The semiconductor substrate processing apparatus of claim 1, wherein the load lock is configured to receive the semiconductor substrate, based on the semiconductor substrate being transferred from the substrate transfer module to the transfer chamber or transferred from the transfer chamber to the substrate transfer module.
10. The semiconductor substrate processing apparatus of claim 9, wherein the load lock includes one or more inner surfaces at least partially defining an internal region inside the load lock, the load lock is configured to accommodate the semiconductor substrate within the internal region, and the load lock is configured to control a pressure of the internal region.
11. The semiconductor substrate processing apparatus of claim 1, wherein the substrate transfer module further includes at least one pair of second load ports on the opposite sides of the chamber so such that the at least one pair of second load ports do not overlap the at least one pair of first load ports in the first direction.
12. The semiconductor substrate processing apparatus of claim 1, wherein the substrate transfer module further includes an upper buffer device in an upper portion of the internal space and configured to store the semiconductor substrate.
13. A semiconductor substrate processing apparatus, comprising: a substrate transfer module, the substrate transfer module including a chamber having one or more inner surfaces at least partially defining an internal space extending in a first direction within the chamber, at least one pair of load ports at opposite sides of the chamber, the at least one pair of load ports facing in a second direction, intersecting the first direction, at least one pair of substrate aligners and at least one pair of buffer devices on opposite side portions of the internal space to overlap the at least one pair of load ports in the first direction, and a robot arm configured to move in the first direction in the internal space of the chamber and configured to transfer a semiconductor substrate to the at least one pair of load ports, the at least one pair of substrate aligners and the at least one pair of buffer devices; a transfer chamber connected to the substrate transfer module; a plurality of processing chambers connected to the transfer chamber; and a transfer arm inside the transfer chamber and configured to load the semiconductor substrate into at least one of the plurality of processing

chambers.

14. The semiconductor substrate processing apparatus of claim 13, wherein the substrate transfer module further includes a load lock connecting the chamber and the transfer chamber.

15. The semiconductor substrate processing apparatus of claim 14, wherein the robot arm is configured to load the semiconductor substrate into the load lock based on the semiconductor substrate being unloaded from the at least one pair of buffer devices, or to load the semiconductor substrate into the at least one pair of buffer devices based on the semiconductor substrate being unloaded from the load lock.

16. The semiconductor substrate processing apparatus of claim 13, wherein the at least one pair of load ports include a transfer tray configured to rotate a substrate carrier about a rotation axis extending in the first direction to connect the substrate carrier accommodating the semiconductor substrate to the chamber.

17. The semiconductor substrate processing apparatus of claim 16, wherein the at least one pair of load ports are configured to rotate the substrate carrier such that a door of the substrate carrier faces the chamber.

18. A semiconductor substrate processing apparatus, comprising: a substrate transfer module, the substrate transfer module including a chamber having a front surface and a rear surface opposing each other, at least one pair of load ports at opposite sides of the chamber, at least one pair of buffer devices in the chamber to overlap the at least one pair of load ports in a vertical direction, a load lock on the rear surface of the chamber, and a robot arm configured to move in the vertical direction in the chamber and configured to transfer a semiconductor substrate to the at least one pair of load ports, the at least one pair of buffer devices, and the load lock, wherein the front surface of the chamber includes an entrance configured to enable the robot arm and the at least one pair of buffer devices to be loaded and unloaded through the entrance.

19. The semiconductor substrate processing apparatus of claim 18, wherein the robot arm and the at least one pair of buffer devices are installed in the chamber and are configured to slide through the entrance.

20. The semiconductor substrate processing apparatus of claim 18, wherein the substrate transfer module further includes at least one front door configured to open and close the entrance.
